



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Generative AI for Agile Project Management

Course Code: C324

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
3.0	1 January 2026	Legacy master trainer deck and lesson plan (v3) — 4 topics, exercise-based delivery.	Dr. Alfred Ang
4.0	17 August 2026	Major revision. Content beefed up from Thoughtworks, Adaptovate, BMC, PMI, UX Magazine, The Design Gym, PremierAgile and Startup Frontier. Twelve real-world case-study activities (Airbnb, Netflix/Blockbuster, GE Adventure Series, IDEO, Spotify, IBM, DBS, Boeing 737 MAX) replace the generic exercises, each with scenario, discussion questions and debrief. Design Thinking Studio and Padlet Classroom Board adopted as the class ed-tools.	Dr. Alfred Ang

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Course Information

Course Title	Generative AI for Agile Project Management
Course Reference	C324
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 instructional hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch break)
Mode	Instructor-led, case-study based, with collaborative online ed-tools
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Integrate design thinking methodologies and agile principles to drive organisational innovation using generative AI.
- LO2: Synthesise stakeholder inputs to uncover end-user needs for successful innovation.
- LO3: Lead design thinking projects using project management tools and techniques to enhance organisational performance.
- LO4: Develop strategies for agile design thinking to enhance product and service innovation.

Skills Framework Alignment

Abilities:

Knowledge:

Teaching and Learning Methods

- Trainer-led concept delivery using a highly visual slide deck (diagrams, comparison tables and frameworks rather than bullet walls).
- Twelve real-world case-study activities, each carrying a documented case, a workplace scenario, five discussion questions and a trainer debrief.
- Collaborative group work in groups of 3–5 using two online ed-tools: the Design Thinking Studio (5-stage workspace) and the Padlet Classroom Board.
- Guided use of generative AI tools (ChatGPT / Microsoft Copilot) with explicit critique of AI output against evidence.
- Cross-group peer review — groups critique each other's output and defend their reasoning.
- Continuous formative feedback from the trainer during each activity, and a structured debrief after each one.

Learning Resources

- Trainer Slides (PPT/PDF), Learner Guide (PDF) and this Lesson Plan (PDF) — downloadable from the LMS at <https://lms-tms.tertiaryinfotech.com>.
- Twelve individual activity briefs (PDF), one per activity, each containing the case, the scenario, the discussion questions, the step-by-step instructions and the debrief.
- Design Thinking Studio — <https://alfredang.github.io/designthinking/>
- Padlet Classroom Board — <https://alfredang.github.io/padlet/>
- A generative AI assistant (ChatGPT or Microsoft Copilot) and a laptop with a modern web browser.

Course Schedule

Day 1 — Foundations, Problem Framing and Ideation with GenAI

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, learning outcomes and course outline. Introduction to the two course ed-tools: Design Thinking Studio and the Padlet Classroom Board.
10:00–11:00	60 min	Topic 1 — Foundations of Design Thinking, Agile and Generative AI: what design thinking is; the three mindsets (Design Thinking, Lean, Agile) and the question each answers; the Double Diamond; the five stages; wicked problems and why framing comes first.
11:00–11:45	45 min	Topic 1 continued — Product development through design thinking and agile; how generative AI enhances the loop; where AI must not replace human judgement; latest trends in innovation.
11:45–12:30	45 min	Hands-on case study: Activity 1: Airbnb — Diagnosing Why a Method Alone Does Not Save a Business

12:30–13:00	30 min	Lunch break
13:00–14:30	90 min	Hands-on case studies: Activity 2: Netflix vs Blockbuster — Innovation as an Operating System; Activity 3: Positioning GenAI in the Innovation Loop — Where It Helps and Where It Lies
14:30–15:30	60 min	Topic 2 — Problem Framing and Ideation: framing the design challenge; the Problem-Assumption Model; problem statement vs Point of View vs How Might We; the altitude test; empathy maps and persona mapping; ideation techniques; prototyping to think.
15:30–17:15	105 min	Hands-on case studies: Activity 4: GE Adventure Series — Reframing the Problem Instead of Fixing the Machine; Activity 5: IDEO Shopping Cart — Volume, Then Judgement
17:15–17:30	15 min	Day 1 recap, Q&A and close

Total instructional time: 450 minutes (7.5 hours).

Day 2 — Agile Delivery and Scaling Innovation

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and orientation to Day 2.
9:45–10:45	60 min	Topic 2 concluded — prototyping with AI-driven tools and defining a falsifiable test plan. Hands-on case study: Activity 6: Rapid Prototyping with AI — From Concept to Testable Artefact in One Hour
10:45–11:45	60 min	Topic 3 — Agile Development and AI for Rapid Solution Delivery: being agile vs doing agile; the Scrum loop and the

		three accountabilities; dual-track agile; the mini-waterfall anti-pattern; Epic → Feature → Story → Task.
11:45–12:30	45 min	Hands-on case study: Activity 7: Spotify — Reading a Scaling Model Honestly
12:30–13:00	30 min	Lunch break
13:00–14:30	90 min	Topic 3 continued — user stories and the 3 C's, the three-cake release model, story points and Planning Poker, GenAI across the sprint. Hands-on case studies: Activity 8: From Hills to Backlog — Converting Ideas into Stories with GenAI; Activity 9: Running Dual-Track Agile — A Sprint Simulation with AI Assistance
14:30–15:30	60 min	Topic 4 — Scaling and Sustaining Innovations: the three levers that scale innovation; culture and the four principles; stakeholder buy-in; resource management with AI; sensemaking; systems thinking and feedback loops; innovation metrics and KPIs.
15:30–17:15	105 min	Hands-on case studies: Activity 10: DBS Bank — Scaling Innovation Across 25,000 People; Activity 11: Boeing 737 MAX — Systems Thinking and the Cost of Local Optimisation; Activity 12: Innovation Metrics — Designing KPIs That Change a Decision
17:15–17:30	15 min	Course revision, Q&A and close

Total instructional time: 450 minutes (7.5 hours).

Activity Reference (aligned to topics and learning outcomes)

Topic	Weighting	Activities	Ed-tool
Topic 01: Foundations of Design Thinking, Agile, and Generative AI for Problem-Solving	25%	Activity 1, Activity 2, Activity 3	Padlet
Topic 02: Problem Framing and Ideation: Leveraging Design Thinking and Generative AI	25%	Activity 4, Activity 5, Activity 6	Design Thinking Studio
Topic 03: Agile Development and AI for Rapid Solution Delivery	25%	Activity 7, Activity 8, Activity 9	Design Thinking Studio / Padlet
Topic 04: Scaling and Sustaining Innovations with Agile Design Thinking and Generative AI	25%	Activity 10, Activity 11, Activity 12	Design Thinking Studio / Padlet

Detailed Activity Schedule

#	Activity	Duration	Real-world case
1	Airbnb — Diagnosing Why a Method Alone Does Not Save a Business	45 min	Airbnb, 2009...
2	Netflix vs Blockbuster — Innovation as an Operating System	45 min	In 2000 Reed Hastings offered to sell Netflix to Blockbuster for \$50 million...
3	Positioning GenAI in the Innovation Loop — Where It Helps and Where It Lies	40 min	A Singapore bank's innovation team used a large language model to generate customer person...
4	GE Adventure Series — Reframing the Problem Instead of Fixing the Machine	50 min	GE Healthcare's MRI scanners were technically excellent, but designer Doug Dietz watched a...
5	IDEO Shopping Cart — Volume, Then Judgement	50 min	In 1999, ABC's Nightline gave the design firm IDEO five days to redesign the supermarket s...
6	Rapid Prototyping with AI — From Concept to Testable Artefact in One Hour	50 min	A Singapore hospital team needed to test whether a self-service kiosk would cut queue time...
7	Spotify — Reading a Scaling Model	45 min	In 2012 Henrik Kniberg and Anders Ivarsson

	Honestly		published a paper describing how Spotify organi...
8	From Hills to Backlog — Converting Ideas into Stories with GenAI	55 min	IBM runs a two-day Design Thinking and Agile workshop in which Day 1 produces 'Hills' — ou...
9	Running Dual-Track Agile — A Sprint Simulation with AI Assistance	50 min	Marty Cagan describes the most common failure he sees in teams that believe they are agile...
10	DBS Bank — Scaling Innovation Across 25,000 People	50 min	Around 2014 DBS Bank set out to 'Make Banking Joyful' and to become a technology company d...
11	Boeing 737 MAX — Systems Thinking and the Cost of Local Optimisation	50 min	Boeing needed a fast, cheap answer to the Airbus A320neo...
12	Innovation Metrics — Designing KPIs That Change a Decision	45 min	Two innovation programmes reported to the same board...